National Research University Higher School of Economics

Faculty of Computer Science

Bachelor Programme "Data Science and Business Analytics"

Course Project

Development of an Alice Voice Assistant Skill for International Applicants of HSE

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Introduction

Voice assistants such as Alice (Yandex), Marusya (VK), and Salute (Sber) have become an important component of the modern artificial intelligence ecosystem, enabling users to access information through natural voice interaction. In the Russian Federation, Alice occupies a leading position among voice assistants due to its strong support for the Russian language, high-quality speech recognition, and deep integration with the Yandex service infrastructure. The Yandex.Dialogs platform enables the development of so-called "skills," which are server-side applications integrated via webhook that process user requests and return responses within the voice assistant ecosystem.

A particularly important category within the university admission system is international applicants. Unlike domestic applicants, they face additional challenges, including language barriers, time zone differences, and the need to navigate the specific administrative and educational systems of the Russian Federation. Admission-related information is often distributed across multiple official documents and web pages, making it difficult for international applicants to independently interpret and structure the required information.

The Higher School of Economics annually attracts a significant number of international applicants, which increases the workload of the admissions office, especially during peak admission periods. Traditional communication channels—such as the official website, email correspondence, and phone consultations—do not always provide timely and convenient responses for users located outside the Russian Federation.

Under these conditions, the development of a specialized voice assistant skill for Alice, targeted specifically at international applicants of HSE, represents a relevant and practical solution. The use of a Retrieval-Augmented Generation (RAG) architecture, which retrieves relevant fragments of official documents before generating a response, allows the system to preserve factual consistency and minimize the risk of hallucinated or outdated information by grounding answers in official university sources.

Project Goal

The goal of this project is to develop a voice assistant skill titled *"HSE International Applicant"* for the Alice voice assistant, which provides international applicants of the Higher School of Economics with up-to-date and reliable information about the university admission process.

Project Objectives

To achieve the stated goal, the following objectives were defined:

1. Analyze the informational needs of international applicants and identify typical inquiry scenarios.
2. Design a system architecture based on the RAG approach, including a retrieval module, knowledge index, and response generation module.
3. Construct and structure a knowledge base derived from official HSE admission documents and web pages.
4. Perform data preprocessing, including text cleaning, chunk segmentation, and metadata annotation.
5. Implement vector-based document indexing using embedding models.
6. Conduct experimental comparison of different embedding models and evaluate their impact on retrieval quality.
7. Investigate the influence of retrieval parameters (e.g., top-k) on contextual completeness and relevance.
8. Compare the effectiveness of vector-based retrieval and keyword-based search methods.
9. Develop fallback strategies for cases of low-confidence retrieval or missing relevant documents.
10. Evaluate system responses in terms of factual grounding, completeness, robustness against hallucinations, and response latency.
11. Conduct system testing and formulate recommendations for further improvement and scalability.

Literature Review

Voice assistants represent a rapidly developing field within artificial intelligence, combining speech recognition, natural language processing, and speech synthesis technologies. In the educational domain, such systems are increasingly used to automate responses to frequently asked questions and to optimize communication between users and institutions.

With the advancement of large language models (LLMs), the issue of hallucination—generation of factually incorrect or fabricated information—has become a critical challenge. In domains that require strict adherence to official regulations, such as university admissions, such errors are unacceptable. To address this problem, the Retrieval-Augmented Generation (RAG) architecture was proposed. RAG combines a document retrieval component with a generative language model, where the model produces responses conditioned on retrieved evidence from an external knowledge base.

The RAG approach constrains generation to verified document fragments, thereby significantly improving factual grounding and reducing the likelihood of hallucinated content. Recent extensions of RAG include hybrid retrieval (combining vector similarity and keyword filtering), query expansion techniques, and confidence-based fallback strategies. These mechanisms are particularly relevant in high-stakes informational domains, including legal, medical, and academic admission systems.

Within the Russian technological ecosystem, the Yandex.Dialogs platform provides a ready-to-use infrastructure for developing voice assistant skills. Leveraging an existing platform substantially reduces technical complexity compared to building a standalone voice assistant, which would require implementing a complete technology stack (Speech-to-Text, NLP, and Text-to-Speech) as well as dedicated cloud infrastructure.

An analysis of existing informational tools for HSE applicants reveals the absence of a specialized voice-based solution grounded in official university documents. This confirms the relevance of developing a voice assistant skill that integrates Yandex.Dialogs infrastructure with a RAG-based architecture to provide accurate, grounded, and accessible informational support for international applicants of the Higher School of Economics.

1 Experimental Evaluation

1.1 Dataset

The system was evaluated on a manually constructed gold dataset consisting of 50 questions related to admission procedures for international applicants to HSE University.

Each dataset entry contains:

- the user question,
- the identifier of the relevant source document,
- the reference answer (used for manual validation).

The document corpus includes 10 official admission-related documents in `.docx` format. After preprocessing and chunking, the corpus produced 314 text chunks used for indexing.

1.2 Retrieval Configuration

ChromaDB was used as the vector storage and retrieval engine. Dense embeddings were generated using the `sentence-transformers/all-MiniLM-L6-v2` model.

The selected chunking configuration:

- Chunk size: 900 characters
- Overlap: 150 characters

For each query, the retriever returns the top-k most relevant chunks, where $k \in \{1, 3, 5\}$.

1.3 Evaluation Metrics

We use standard retrieval metrics:

- **Hit@k**: equals 1 if the correct document appears among the top-k retrieved results.
- **MRR@k (Mean Reciprocal Rank)**: evaluates how high the correct document appears in the ranking.

Mean Reciprocal Rank is defined as:

$$MRR = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{rank_i}$$

where $rank_i$ denotes the position of the correct document for query i.

1.4 Baseline Retrieval Performance

The baseline configuration (MiniLM embeddings, chunk size 900/150) was evaluated on 50 gold questions.

	Hit@1	Hit@3	Hit@5	MRR@1	MRR@3	MRR@5
Baseline (900/150)	0.20	0.30	0.30	0.20	0.2467	0.2467

Table 1: Baseline dense retrieval performance

1.5 Impact of Chunk Size

Chunk Size / Overlap	Hit@1	Hit@3	Hit@5	MRR@1	MRR@3	MRR@5
600 / 100	0.14	0.22	0.32	0.14	0.1733	0.1963
900 / 150	**0.20**	**0.30**	0.30	**0.20**	**0.2467**	**0.2467**
1200 / 200	0.12	0.20	0.20	0.12	0.1533	0.1803
1500 / 250	0.12	0.14	0.24	0.12	0.1300	0.1500

Table 2: Effect of chunk size on retrieval quality

1.6 Discussion

The results indicate that medium-sized chunks (900/150) provide the best balance between contextual completeness and ranking precision.

Smaller chunks slightly improve Hit@5 but reduce ranking quality (lower MRR), indicating context fragmentation. Larger chunks decrease retrieval accuracy due to semantic dilution.

1.7 Conclusion of Experimental Study

The implemented dense retrieval system demonstrates baseline performance for admission-related queries:

- Hit@1 = 0.20
- Hit@3 = 0.30
- MRR@5 = 0.2467

Future improvements may include testing stronger multilingual embedding models, hybrid retrieval approaches (BM25 + dense retrieval), and reranking techniques to enhance ranking precision.